

MM 1

$$\begin{bmatrix} \text{pink} & \text{pink} \\ \text{pink} & \text{pink} \\ \text{pink} & \text{pink} \end{bmatrix} \begin{bmatrix} \text{green} & \text{green} \\ \text{green} & \text{green} \end{bmatrix} = \begin{bmatrix} \text{green/pink} & \text{green/pink} \\ \text{green/pink} & \text{green/pink} \\ \text{green/pink} & \text{green/pink} \end{bmatrix}$$

Every element becomes a dot product of row vector and column vector.

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} x_1 & y_1 \\ x_2 & y_2 \end{bmatrix} = \begin{bmatrix} (x_1+2x_2) & (y_1+2y_2) \\ (3x_1+4x_2) & (3y_1+4y_2) \\ (5x_1+6x_2) & (5y_1+6y_2) \end{bmatrix}$$

MM 2

$$\begin{bmatrix} \text{gray} & \text{gray} & \text{gray} & \text{gray} \end{bmatrix} \begin{bmatrix} \text{green} & \text{green} \\ \text{green} & \text{green} \end{bmatrix} = \begin{bmatrix} \text{gray} & \text{green} & \text{gray} & \text{green} \\ \text{gray} & \text{green} & \text{gray} & \text{green} \end{bmatrix} = \begin{bmatrix} \text{green} & \text{green} \\ \text{green} & \text{green} \end{bmatrix}$$

$A\mathbf{x}$ and $A\mathbf{y}$ are linear combinations of columns of A .

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} x_1 & y_1 \\ x_2 & y_2 \end{bmatrix} = A \begin{bmatrix} \mathbf{x} & \mathbf{y} \end{bmatrix} = \begin{bmatrix} A\mathbf{x} & A\mathbf{y} \end{bmatrix}$$

MM 3

$$\begin{bmatrix} \text{pink} & \text{pink} \\ \text{pink} & \text{pink} \\ \text{pink} & \text{pink} \end{bmatrix} \begin{bmatrix} \text{gray} & \text{gray} \\ \text{gray} & \text{gray} \end{bmatrix} = \begin{bmatrix} \text{pink/gray} & \text{pink/gray} \\ \text{pink/gray} & \text{pink/gray} \\ \text{pink/gray} & \text{pink/gray} \end{bmatrix} = \begin{bmatrix} \text{pink} & \text{gray} \\ \text{pink} & \text{gray} \\ \text{pink} & \text{gray} \end{bmatrix}$$

The produced rows are linear combinations of rows.

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} x_1 & y_1 \\ x_2 & y_2 \end{bmatrix} = \begin{bmatrix} \mathbf{a}_1^* \\ \mathbf{a}_2^* \\ \mathbf{a}_3^* \end{bmatrix} X = \begin{bmatrix} \mathbf{a}_1^* X \\ \mathbf{a}_2^* X \\ \mathbf{a}_3^* X \end{bmatrix}$$

MM 4

$$\begin{bmatrix} \text{green} & \text{green} \\ \text{green} & \text{green} \end{bmatrix} \begin{bmatrix} \text{pink} & \text{pink} \\ \text{pink} & \text{pink} \end{bmatrix} = \begin{bmatrix} \text{pink/green} & \text{pink/green} \\ \text{pink/green} & \text{pink/green} \end{bmatrix} + \begin{bmatrix} \text{pink/gray} & \text{pink/gray} \\ \text{pink/gray} & \text{pink/gray} \end{bmatrix}$$

Multiplication AB is broken down to a sum of rank 1 matrices.

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 \end{bmatrix} \begin{bmatrix} \mathbf{b}_1^* \\ \mathbf{b}_2^* \end{bmatrix} = \mathbf{a}_1 \mathbf{b}_1^* + \mathbf{a}_2 \mathbf{b}_2^* \\ = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \end{bmatrix} + \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix} \begin{bmatrix} b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} b_{11} & b_{12} \\ 3b_{11} & 3b_{12} \\ 5b_{11} & 5b_{12} \end{bmatrix} + \begin{bmatrix} 2b_{21} & 2b_{22} \\ 4b_{21} & 4b_{22} \\ 6b_{21} & 6b_{22} \end{bmatrix}$$